

ZeroDrift MK2: shopping list

Team ZeroDrift · Jabalpur Engineering College · SIH 2026 · for Amar Robotics

For the shopkeeper: we are building a small pan-tilt camera turret with two **NEMA17 stepper motors (5 mm shaft)** and **A4988 drivers**, run by an Arduino Nano. We need the items below.

Remember: the B3 charger never connects to the rig; it only charges batteries. The 5 V 1 A adapter can't run the motors (it's for the beacon). The motors run on 12 V: item 2, or charged 18650s (3.7–4.2 V each).

A · Motors, power and electronics

NO.	ITEM (ASK FOR)	QTY	DESCRIPTION FOR THE SHOPKEEPER · WHAT IT IS FOR	≈ ₹
☐ 1	5 mm rigid flange coupling	2	Metal flange coupling with a 5 mm bore , to fit a NEMA17 stepper motor shaft. Rigid (solid) type, not flexible or spring. Grub screw, and a flange with 4 screw holes. Holds the pan platform and the tilt disc on the two motor shafts.	160–300
☐ 2	12 V 2 A DC adapter (SMPS)	1	Wall adapter, 12 V DC, 2 A , 5.5 × 2.1 mm barrel plug. Powers the two stepper motors. (Our 5 V 1 A adapter is too weak for motors.)	200–350
☐ 3	DC female jack with screw terminals	1	5.5 × 2.1 mm, to match the adapter, with + and – screw terminals. Connects the adapter to the breadboard wires.	20–40
☐ 4	Resistor 100 Ω, ¼ W	5	Colour bands brown-black-brown-gold . For the decoy LED.	5–10
☐ 5	Resistor 220 Ω, ¼ W	5	Colour bands red-red-brown-gold . For the beacon LED.	5–10
☐ 6	Red LED, high brightness, 10 mm	2	Bright red LED (5 mm also fine). One for the decoy, one spare.	10–20

B · Frame and holders (platform, discs, the two encoder arms, head)

NO.	ITEM (ASK FOR)	QTY	DESCRIPTION FOR THE SHOPKEEPER · WHAT IT IS FOR	≈ ₹
☐ 7	Clear acrylic sheet, 3 mm, A4 size	1	Transparent acrylic sheet. We cut from it: the pan platform disc (6 cm), the tilt disc (4 cm), the two encoder arms (strips 1.5 × 6 cm, the "yellow parts") and the head plate . Stiff, so nothing bends.	150–250
☐ 8	Acrylic cutter (hook blade)	1	Scoring knife for cutting acrylic: score 5–6 times along a ruler, then snap. A normal paper cutter will not cut 3 mm acrylic.	60–120

☐	9	M3 × 40 mm screw + nuts	4	4 screws, 12 nuts. Standing posts for the two encoder arms (the yellow parts). The nuts set the height.	20–40
☐	10	M3 standoff, 15 mm (brass or nylon)	4	Raise the tilt motor's L-bracket above the pan encoder arm.	30–60
☐	11	M3 brass standoff, 20 mm (male–female)	4	Screwed together in pairs = 40 mm. This is the post the camera head hangs from. Brass, not plastic.	40–80
☐	12	M3 screws + nuts + washers	1 set	M3 × 16 mm × 8 (pan motor through the base, with spacer nuts), M3 × 10 mm × 20 (tilt motor, couplings to discs, L-bracket), 30 nuts, 20 washers. NEMA17 motors take M3 only.	60–100
☐	13	M4 × 80 mm bolt + nuts (base legs)	4	4 bolts, 12 nuts. Legs at the 4 corners of the base plate, so the pan motor hangs free under it. The nuts make them level.	30–60
☐	14	Allen key set, small (1.5–3 mm)	1	For the grub screws in the flange couplings (skip if the couplings come with a key).	60–120

C · Boxes, filter and diffuser (beacon, decoy, camera)

NO.	ITEM (ASK FOR)	QTY	DESCRIPTION FOR THE SHOPKEEPER · WHAT IT IS FOR	≈ ₹
☐	15 ABS plastic project box, ≈ 100 × 60 × 25 mm	2	Small plastic enclosure with a lid. One for the beacon (Nano + LED), one for the decoy (3×AA holder + LED + switch).	160–300
☐	16 ABS plastic project box, ≈ 70 × 50 × 30 mm	1	Small box for the camera head : the webcam board and the laser go inside, looking out through two holes in the front.	60–120
☐	17 Red transparent acrylic, 2 mm, small piece (≈ 5 × 5 cm)	1	Or red cellophane / red gel sheet. Goes over the camera lens so bright rooms don't blind it. Check: through it the room looks dim red.	20–80
☐	18 AA batteries (1.5 V)	6	3 in the decoy holder, 3 spare. Alkaline (not "heavy duty") so the decoy stays bright all night.	60–120

D · Glue, tape and small consumables

NO.	ITEM (ASK FOR)	QTY	DESCRIPTION FOR THE SHOPKEEPER · WHAT IT IS FOR	≈ ₹
☐	19 Fevikwik (superglue)	2	Glues the two magnets exactly on the disc centres.	20–40
☐	20 Hot glue sticks (small)	5	For the glue gun: LEDs in the lids, camera and laser in the head, switches.	20–40

☐	21 Double-sided foam tape	1	Sticks down the breadboard, the battery holder and the AS5600 boards.	40–80
☐	22 Insulation tape (black + red)	2	Covers the soldered LED and resistor joints; red marks the + wires.	20–40
☐	23 Zip ties, 100 mm	1	1 packet. Ties the cables with a loose loop so the rig can turn.	30–50
☐	24 Solder wire, small reel (skip if you have)	1	For the header pins on the TCA9548A and AS5600 boards and the LED + resistor joints.	40–80
Total for A + B + C + D				≈ ₹1,300–2,400

E · Only if needed

NO.	ITEM (ASK FOR)	QTY	BUY IT ONLY WHEN...	≈ ₹
☐	25 TP4056 charging module	1	Only if the shop has no 12 V adapter (item 2): then we charge our 18650 cells one at a time with it and run the motors from them.	30–60
☐	26 USB cable for Arduino Nano	1	Only if we don't have one matching the second Nano's port (Mini-USB or USB-C).	60–120
☐	27 USB-C hub (USB-C to 2× USB-A)	1	Only if the laptop has USB-C ports only: the Nano and the webcam must be plugged in together.	300–800

Already have: don't buy

- ✓ 2× NEMA17 stepper motors
- ✓ 2× NEMA17 L-brackets
- ✓ 2× AS5600 encoders + diametric magnets
- ✓ 2× Arduino Nano
- ✓ Red and white LEDs
- ✓ Ping-pong ball
- ✓ 5 V 1 A adapter (for the beacon)
- ✓ Brass standoffs, M3 screws and nuts
- ✓ Multimeter, soldering iron, glue gun
- ✓ USB webcam (arriving)
- ✓ 2× A4988 drivers (with heatsinks)
- ✓ 100 µF capacitors
- ✓ TCA9548A multiplexer
- ✓ KY-008 laser
- ✓ 3×AA battery holder
- ✓ 3× 18650 cells + holder, B3 charger
- ✓ Toggle switches
- ✓ Breadboard + jumper wires
- ✓ White sheet (base plate)
- ✓ Foam tape, tape, scissors

From home

- △ Ruler, marker, scissors
- △ Phone USB charger or power bank: powers the beacon Nano
- △ Two old credit cards (a 1.5 mm gap gauge for the encoders)
- △ Glue gun, soldering iron, multimeter
- △ Tiny screwdriver for the A4988 Vref pot
- △ A drill, or someone who has one (about 12 holes)